JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Binomial Theorem

JEE Main Mathematics · Previous Year Questions with Answer Key

255	179	2002 – 2026	70	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	117 questions
Numerical	68 questions
Out of syllabus	70 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 117 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

If $26\left(\frac{2^3}{3}\left({}^{12}C_2\right)+\frac{2^5}{5}\left({}^{12}C_4\right)+\frac{2^7}{7}\left({}^{12}C_6\right)+\cdots+\frac{2^{13}}{13}\left({}^{12}C_{12}\right)\right)=3^{13}-\alpha$, then α is equal to :

(A) 45 (B) 48 (C) 51 (D) 54

Q2. JEE Main 2026 (Online) 6th April Morning Shift

If the coefficients of the middle terms in the binomial expansions of $(1+\alpha x)^{26}$ and $(1-\alpha x)^{28}, \alpha \neq 0$, are equal, then the value of α is:

(A) 1 (B) $\frac{14}{13}$ (C) $\frac{27}{7}$ (D) $\frac{7}{27}$

Q3. JEE Main 2026 (Online) 5th April Evening Shift

The coefficient of x^2 in the expansion of $\left(2x^2+\frac{1}{x}\right)^{10}, x \neq 0$, is :

(A) 3240 (B) 3360 (C) 3480 (D) 3600

Q4. JEE Main 2026 (Online) 4th April Evening Shift

In the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}, x>0$, if the term independent of x is (221)k, then k is equal to:

(A) 84 (B) 78 (C) 168 (D) 198

Q5. JEE Main 2026 (Online) 4th April Morning Shift

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3+(1+x)^4+(1+x)^5+\ldots+(1+x)^{99}+(1+kx)^{100}, x \neq 0$, is $\left(43n+\frac{101}{4}\right)\left({}^{100}C_3\right)$ for some $n \in \mathbb{N}$, be p. Then the value of $p+n$ is :

(A) 10 (B) 11 (C) 12 (D) 13

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

If for $3 \leq r \leq 30$, $\left({}^{30}C_{30-r}\right)+3\left({}^{30}C_{31-r}\right)+3\left({}^{30}C_{32-r}\right)+\left({}^{30}C_{33-r}\right)={}^{m}C_r$, then m equals :

(A) 31 (B) 32 (C) 33 (D) 34

Q7. JEE Main 2026 (Online) 28th January Evening Shift

Given below are two statements :

Statement I :

$25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II :

The integral part of $(7 + 4\sqrt{3})^{25}$ is an odd number.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) **Statement I** is false but **Statement II** is true

(B) Both **Statement I** and **Statement II** are false

(C) Both **Statement I** and **Statement II** are true

(D) **Statement I** is true but **Statement II** is false

Q8. JEE Main 2026 (Online) 28th January Evening Shift

The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots + x^{1000}$ is :

(A) ${}^{1002}C_{501}$ (B) ${}^{1001}C_{501}$ (C) ${}^{1000}C_{501}$ (D) ${}^{1002}C_{500}$

Q9. JEE Main 2026 (Online) 24th January Morning Shift

Let $\mathrm{S} = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots$ up to 13 terms. If $13\,\mathrm{S} = \frac{2^k}{n!}, k \in \mathrm{N}$, then $n + k$ is equal to

(A) 50 (B) 52 (C) 49 (D) 51

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

The sum of all possible values of $\mathbf{n} \in \mathbf{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $\left(1+x^2\right)^2(1+x)^{\mathrm{n}}$, are in arithmetic progression is :

(A) 12 (B) 9 (C) 3 (D) 7

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

The value of $\frac{{}^{100}\mathrm{C}_{50}}{51} + \frac{{}^{100}\mathrm{C}_{51}}{52} + \ldots . + \frac{{}^{100}\mathrm{C}_{100}}{101}$ is:

(A) $\frac{2^{101}}{101}$ (B) $\frac{2^{100}}{101}$ (C) $\frac{2^{100}}{100}$ (D) $\frac{2^{101}}{100}$

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\mathrm{C_r}$ denote the coefficient of x^{r} in the binomial expansion of $(1+x)^{\mathrm{n}}, \mathrm{n} \in \mathrm{N}, 0 \leq \mathrm{r} \leq \mathrm{n}$. If

$P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + \ldots . + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals.

(A) 675 (B) 580 (C) 525 (D) 650

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

The coefficient of x^{48} in $(1+x)+2(1+x)^2+3(1+x)^3+\ldots+100(1+x)^{100}$ is equal to

(A) $100 \cdot {}^{100}C_{49} - {}^{100}C_{48}$ (B) $100 \cdot {}^{101}C_{49} - {}^{101}C_{50}$

(C) ${}^{100}C_{50} + {}^{101}C_{49}$ (D) $100 \cdot {}^{100}C_{49} - {}^{100}C_{50}$

Q14. JEE Main 2026 (Online) 21st January Morning Shift

If the coefficient of x in the expansion of $(ax^2+bx+c)(1-2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to :

(A) 1483 (B) 1300 (C) 1500 (D) 1403

Q15. JEE Main 2025 (Online) 8th April Evening Shift

The number of integral terms in the expansion of $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1016}$ is:

(A) 127 (B) 128 (C) 130 (D) 129

Q16. JEE Main 2025 (Online) 7th April Morning Shift

The remainder when $\left((64)^{(64)}\right)^{(64)}$ is divided by 7 is equal to

(A) 4 (B) 6 (C) 3 (D) 1

Q17. JEE Main 2025 (Online) 4th April Evening Shift

If $1^2 \cdot \left({}^{15}C_1\right) + 2^2 \cdot \left({}^{15}C_2\right) + 3^2 \cdot \left({}^{15}C_3\right) + \ldots + 15^2 \cdot \left({}^{15}C_{15}\right) = 2^m \cdot 3^n \cdot 5^k$, where $m, n, k \in \mathbf{N}$, then $\mathrm{m}+\mathrm{n}+\mathrm{k}$ is equal to :

(A) 20 (B) 19 (C) 18 (D) 21

Q18. JEE Main 2025 (Online) 4th April Morning Shift

For an integer $n \geq 2$, if the arithmetic mean of all coefficients in the binomial expansion of $(x+y)^{2n-3}$ is 16 , then the distance of the point $\mathrm{P}\left(2n-1, n^2-4n\right)$ from the line $x+y=8$ is

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) $5\sqrt{2}$ (D) $3\sqrt{2}$

Q19. JEE Main 2025 (Online) 4th April Morning Shift

In the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n, n \in \mathrm{N}$, if the ratio of 15th term from the beginning to the 15th term from the end is $\frac{1}{6}$, then the value of ${}^{n}C_3$ is

(A) 4960 (B) 2300 (C) 1040 (D) 4060

Q20. JEE Main 2025 (Online) 3rd April Morning Shift

The sum of all rational terms in the expansion of $(2+\sqrt{3})^8$ is :

(A) 16923 (B) 18817 (C) 3763 (D) 33845

Q21. JEE Main 2025 (Online) 3rd April Morning Shift

If $\sum_{r=1}^{9}\left(\frac{r+3}{2^r}\right)\cdot {}^9C_r = \alpha\left(\frac{3}{2}\right)^9 - \beta, \alpha, \beta \in \mathbb{N}$, then $(\alpha+\beta)^2$ is equal to

(A) 27 (B) 81 (C) 18 (D) 9

Q22. JEE Main 2025 (Online) 2nd April Evening Shift

$If \sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right).{}^{11}C_{r+1} = \frac{\alpha^{11}-11^{11}}{10^{10}}, then\ \alpha\ is\ equal\ to$:

(A) 11 (B) 20 (C) 24 (D) 15

Q23. JEE Main 2025 (Online) 2nd April Morning Shift

The largest $\mathrm{n} \in \mathbf{N}$ such that 3^{n} divides 50 ! is :

(A) 22 (B) 20 (C) 21 (D) 23

Q24. JEE Main 2025 (Online) 2nd April Morning Shift

The term independent of x in the expansion of $\left(\frac{(x+1)}{\left(x^{2/3}+1-x^{1/3}\right)}-\frac{(x-1)}{\left(x-x^{1/2}\right)}\right)^{10}, x>1$, is :

(A) 240 (B) 120 (C) 150 (D) 210

Q25. JEE Main 2025 (Online) 29th January Evening Shift

The remainder, when 7^{103} is divided by 23, is equal to:

(A) 9 (B) 6 (C) 14 (D) 17

Q26. JEE Main 2025 (Online) 29th January Morning Shift

The least value of **n** for which the number of integral terms in the Binomial expansion of $(\sqrt[3]{7}+\sqrt[12]{11})^n$ is 183, is :

(A) 2184 (B) 2172 (C) 2196 (D) 2148

Q27. JEE Main 2025 (Online) 28th January Evening Shift

Let the coefficients of three consecutive terms T_r, T_{r+1} and T_{r+2} in the binomial expansion of $(a+b)^{12}$ be in a G.P. and let p be the number of all possible values of r. Let q be the sum of all rational terms in the binomial expansion of $(\sqrt[4]{3}+\sqrt[3]{4})^{12}$. Then $p+q$ is equal to:

(A) 295 (B) 283 (C) 299 (D) 287

Q28. JEE Main 2025 (Online) 24th January Evening Shift

Suppose A and B are the coefficients of 30th and 12th terms respectively in the binomial expansion of $(1+x)^{2n-1}$. If $2\,A = 5\,B$, then n is equal to:

(A) 20 (B) 19 (C) 22 (D) 21

Q29. JEE Main 2025 (Online) 24th January Morning Shift

For some $n \neq 10$, let the coefficients of the 5 th, 6 th and 7 th terms in the binomial expansion of $(1+x)^{n+4}$ be in A.P. Then the largest coefficient in the expansion of $(1+x)^{n+4}$ is:

(A) 10 (B) 35 (C) 70 (D) 20

Q30. JEE Main 2025 (Online) 23rd January Evening Shift

If in the expansion of $(1+x)^p(1-x)^q$, the coefficients of x and x^2 are 1 and -2 , respectively, then p^2+q^2 is equal to :

(A) 8 (B) 20 (C) 13 (D) 18

Q31. JEE Main 2025 (Online) 22nd January Evening Shift

Let α, β, γ and δ be the coefficients of x^7, x^5, x^3 and x respectively in the expansion of $\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, x>1$. If u and v satisfy the equations

$\alpha u+\beta v=18,$

$\gamma u+\delta v=20,$

then $u+v$ equals :

(A) 4 (B) 3 (C) 5 (D) 8

Q32. JEE Main 2024 (Online) 9th April Evening Shift

The sum of the coefficients of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3}+\frac{1}{2}x^{-2/5}\right)^9$ is

(A) 19/4 (B) 69/16 (C) 63/16 (D) 21/4

Q33. JEE Main 2024 (Online) 9th April Morning Shift

The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+\ldots+x^{54}(1+x)^{46}$ is ${}^{99}\mathrm{C}_{\mathrm{p}}-{}^{46}\mathrm{C}_{\mathrm{q}}$. Then a possible value of $\mathrm{p}+\mathrm{q}$ is :

(A) 61 (B) 83 (C) 55 (D) 68

Q34. JEE Main 2024 (Online) 8th April Evening Shift

If the term independent of x in the expansion of $\left(\sqrt{\mathrm{a}}x^2+\frac{1}{2x^3}\right)^{10}$ is 105 , then a^2 is equal to :

(A) 6 (B) 4 (C) 2 (D) 9

Q35. JEE Main 2024 (Online) 5th April Evening Shift

If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}, x\neq 0$, is $\alpha\times 2^8\times\sqrt[5]{3}$, then 25α is equal to :

(A) 724 (B) 742 (C) 693 (D) 639

Q36. JEE Main 2024 (Online) 4th April Evening Shift

If the coefficients of x^4, x^5 and x^6 in the expansion of $(1+x)^n$ are in the arithmetic progression, then the maximum value of n is:

(A) 28 (B) 21 (C) 7 (D) 14

Q37. JEE Main 2024 (Online) 4th April Morning Shift

The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is equal to :

(A) 633 (B) 6131 (C) 3133 (D) 931

Q38. JEE Main 2024 (Online) 1st February Evening Shift

Let m and n be the coefficients of seventh and thirteenth terms respectively

in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}}+\frac{1}{2x^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{\mathrm{n}}{\mathrm{m}}\right)^{\frac{1}{3}}$ is :

(A) $\frac{1}{9}$ (B) $\frac{1}{4}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$

Q39. JEE Main 2024 (Online) 31st January Morning Shift

Let a be the sum of all coefficients in the expansion of $\left(1-2x+2x^2\right)^{2023}\left(3-4x^2+2x^3\right)^{2024}$ and $b=\lim\limits_{x\to 0}\left(\frac{\int_0^x\frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equation $cx^2+dx+e=0$ and $2bx^2+ax+4=0$ have a common root, where $c,d,e\in\mathbb{R}$, then $\mathrm{d}:\mathrm{c}:\mathrm{e}$ equals

(A) $2:1:4$ (B) $1:1:4$ (C) $1:2:4$ (D) $4:1:4$

Q40. JEE Main 2024 (Online) 30th January Evening Shift

Suppose $2-p, p, 2-\alpha, \alpha$ are the coefficients of four consecutive terms in the expansion of $(1+x)^n$. Then the value of $p^2-\alpha^2+6\alpha+2p$ equals

(A) 8 (B) 4 (C) 6 (D) 10

Q41. JEE Main 2024 (Online) 27th January Morning Shift

${}^{n-1}C_r = \left(k^2-8\right){}^nC_{r+1}$ if and only if :

(A) $2\sqrt{2} < \mathrm{k} < 2\sqrt{3}$ (B) $2\sqrt{2} < \mathrm{k} \leq 3$

(C) $2\sqrt{3} < \mathrm{k} < 3\sqrt{3}$ (D) $2\sqrt{3} < \mathrm{k} \leq 3\sqrt{2}$

Q42. JEE Main 2024 (Online) 27th January Morning Shift

If A denotes the sum of all the coefficients in the expansion of $\left(1-3x+10x^2\right)^{\mathrm{n}}$ and B denotes the sum of all the coefficients in the expansion of $\left(1+x^2\right)^n$, then :

(A) $\mathrm{B} = \mathrm{A}^3$ (B) $3\mathrm{A} = \mathrm{B}$ (C) $A = 3B$ (D) $\mathrm{A} = \mathrm{B}^3$

Q43. JEE Main 2023 (Online) 15th April Morning Shift

Let $\left(a+bx+cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$.

If $p_1 = 20$ and $p_2 = 210$, then $2(a+b+c)$ is equal to :

(A) 15 (B) 8 (C) 6 (D) 12

Q44. JEE Main 2023 (Online) 13th April Evening Shift

The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is :

(A) $\frac{26}{3}$ (B) $\frac{80}{9}$ (C) 9 (D) 8

Q45. JEE Main 2023 (Online) 13th April Morning Shift

Fractional part of the number $\frac{4^{2022}}{15}$ is equal to

(A) $\frac{8}{15}$ (B) $\frac{4}{15}$ (C) $\frac{1}{15}$ (D) $\frac{14}{15}$

Q46. JEE Main 2023 (Online) 11th April Evening Shift

The sum of the coefficients of three consecutive terms in the binomial expansion of $(1+\mathrm{x})^{\mathrm{n}+2}$, which are in the ratio $1:3:5$, is equal to :

(A) 63 (B) 92 (C) 25 (D) 41

Q47. JEE Main 2023 (Online) 11th April Evening Shift

If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then $|x|$ is equal to

(A) $\frac{5}{16}$ (B) 8 (C) 12 (D) 15

Q48. JEE Main 2023 (Online) 10th April Evening Shift

Let the number $(22)^{2022}+(2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $\left(\alpha^2+\beta^2\right)$ is equal to :

(A) 13 (B) 10 (C) 20 (D) 5

Q49. JEE Main 2023 (Online) 10th April Evening Shift

If the coefficients of x and x^2 in $(1+x)^{\mathrm{p}}(1-x)^{\mathrm{q}}$ are 4 and -5 respectively, then $2p+3q$ is equal to :

(A) 66 (B) 60 (C) 69 (D) 63

Q50. JEE Main 2023 (Online) 10th April Morning Shift

If the coefficient of x^7 in $\left(ax-\frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax+\frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to :

(A) 22 (B) 33 (C) 44 (D) 11

Q51. JEE Main 2023 (Online) 8th April Evening Shift

$25^{190}-19^{190}-8^{190}+2^{190}$ is divisible by :

(A) 14 but not by 34 (B) neither 14 nor 34

(C) both 14 and 34 (D) 34 but not by 14

Q52. JEE Main 2023 (Online) 8th April Evening Shift

The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2+\frac{1}{2x}\right)^{11}$ is equal to :

(A) 11^3-11 (B) 13^3-13 (C) 12^3-12 (D) 10^3-10

Q53. JEE Main 2023 (Online) 6th April Evening Shift

If the coefficient of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$ are equal, then :

(A) $243ab=64$ (B) $32ab=729$ (C) $64ab=243$ (D) $729ab=32$

Q54. JEE Main 2023 (Online) 6th April Evening Shift

Among the statements :

(S1) : $2023^{2022} - 1999^{2022}$ is divisible by 8

(S2) : $13(13)^n - 12n - 13$ is divisible by 144 for infinitely many $n \in \mathbb{N}$

(A) both (S1) and (S2) are incorrect

(B) only (S1) is correct

(C) only (S2) is correct

(D) both (S1) and (S2) are correct

Q55. JEE Main 2023 (Online) 6th April Morning Shift

If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6} : 1$, then the third term from the beginning is :

(A) $30\sqrt{2}$ (B) $60\sqrt{3}$ (C) $60\sqrt{2}$ (D) $30\sqrt{3}$

Q56. JEE Main 2023 (Online) 30th January Evening Shift

Let $x = (8\sqrt{3} + 13)^{13}$ and $y = (7\sqrt{2} + 9)^9$. If $[t]$ denotes the greatest integer $\leq t$, then :

(A) $[x]$ is odd but $[y]$ is even

(B) $[x]$ and $[y]$ are both odd

(C) $[x] + [y]$ is even

(D) $[x]$ is even but $[y]$ is odd

Q57. JEE Main 2023 (Online) 30th January Morning Shift

If the coefficient of x^{15} in the expansion of $\left(\mathrm{a}x^3 + \frac{1}{\mathrm{b}x^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

(A) a = 3b (B) ab = 1 (C) ab = 3 (D) a = b

Q58. JEE Main 2023 (Online) 29th January Evening Shift

Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1 + x)^{99}$. Let a be the middle term in the expansion of $\left(2 + \frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{^{200}C_{99}K}{a} = \frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

(A) (50, 101) (B) (50, 51) (C) (51, 101) (D) (51, 99)

Q59. JEE Main 2023 (Online) 25th January Evening Shift

$\sum_{k=0}^{6} {}^{51-k}C_3$ is equal to :

(A) ${}^{51}\mathrm{C}_4 - {}^{45}\mathrm{C}_4$

(B) ${}^{51}\mathrm{C}_3 - {}^{45}\mathrm{C}_3$

(C) ${}^{52}\mathrm{C}_3 - {}^{45}\mathrm{C}_3$

(D) ${}^{52}\mathrm{C}_4 - {}^{45}\mathrm{C}_4$

Q60. JEE Main 2022 (Online) 28th July Morning Shift

The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is :

(A) 0 (B) 2 (C) 3 (D) 4

Q61. JEE Main 2022 (Online) 27th July Morning Shift

The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is

(A) 0 (B) 1 (C) 2 (D) 6

Q62. JEE Main 2022 (Online) 25th July Evening Shift

The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is

(A) 1 (B) 4 (C) 6 (D) 8

Q63. JEE Main 2022 (Online) 30th June Morning Shift

For two positive real numbers a and b such that $\frac{1}{a^2} + \frac{1}{b^3} = 4$, then minimum value of the constant term in the expansion of $\left(ax^{\frac{1}{8}} + bx^{-\frac{1}{12}}\right)^{10}$ is :

(A) $\frac{105}{2}$ (B) $\frac{105}{4}$ (C) $\frac{105}{8}$ (D) $\frac{105}{16}$

Q64. JEE Main 2022 (Online) 29th June Morning Shift

If the constant term in the expansion of

$\left(3x^3 - 2x^2 + \frac{5}{x^5}\right)^{10}$ is $2^k.l$, where l is an odd integer, then the value of k is equal to:

(A) 6 (B) 7 (C) 8 (D) 9

Q65. JEE Main 2022 (Online) 28th June Evening Shift

The term independent of x in the expansion of

$(1 - x^2 + 3x^3)\left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}$, $x \neq 0$ is :

(A) $\frac{7}{40}$ (B) $\frac{33}{200}$ (C) $\frac{39}{200}$ (D) $\frac{11}{50}$

Q66. JEE Main 2022 (Online) 26th June Morning Shift

The remainder when $(2021)^{2023}$ is divided by 7 is :

(A) 1 (B) 2 (C) 5 (D) 6

Q67. JEE Main 2022 (Online) 25th June Evening Shift

The coefficient of x^{101} in the expression $(5+x)^{500}+x(5+x)^{499}+x^2(5+x)^{498}+\ldots..+x^{500}$, x > 0, is

(A) $^{501}C_{101}\,(5)^{399}$ (B) $^{501}C_{101}\,(5)^{400}$ (C) $^{501}C_{100}\,(5)^{400}$ (D) $^{500}C_{101}\,(5)^{399}$

Q68. JEE Main 2022 (Online) 24th June Morning Shift

The remainder when 3^{2022} is divided by 5 is :

(A) 1 (B) 2 (C) 3 (D) 4

Q69. JEE Main 2021 (Online) 27th July Evening Shift

A possible value of 'x', for which the ninth term in the expansion of $\left\{3^{\log_3\sqrt{25^{x-1}+7}}+3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}\right\}^{10}$ in the increasing powers of $3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}$ is equal to 180, is :

(A) 0 (B) -1 (C) 2 (D) 1

Q70. JEE Main 2021 (Online) 27th July Morning Shift

If the coefficients of x^7 in $\left(x^2+\frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x-\frac{1}{bx^2}\right)^{11}$, $b\neq 0$, are equal, then the value of b is equal to :

(A) 2 (B) -1 (C) 1 (D) -2

Q71. JEE Main 2021 (Online) 25th July Evening Shift

The sum of all those terms which are rational numbers in the expansion of $(2^{1/3}+3^{1/4})^{12}$ is :

(A) 89 (B) 27 (C) 35 (D) 43

Q72. JEE Main 2021 (Online) 25th July Evening Shift

If the greatest value of the term independent of 'x' in the expansion of $\left(x\sin\alpha+a\frac{\cos\alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of 'a' is equal to :

(A) -1 (B) 1 (C) -2 (D) 2

Q73. JEE Main 2021 (Online) 25th July Evening Shift

The lowest integer which is greater than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is _____.

(A) 3 (B) 4 (C) 2 (D) 1

Q74. JEE Main 2021 (Online) 25th July Morning Shift

If b is very small as compared to the value of a, so that the cube and other higher powers of $\frac{b}{a}$ can be neglected in the identity $\frac{1}{a-b}+\frac{1}{a-2b}+\frac{1}{a-3b}+\ldots\ldots+\frac{1}{a-nb}=\alpha n+\beta n^2+\gamma n^3$, then the value of γ is :

(A) $\frac{a^2+b}{3a^3}$ (B) $\frac{a+b}{3a^2}$ (C) $\frac{b^2}{3a^3}$ (D) $\frac{a+b^2}{3a^3}$

Q75. JEE Main 2021 (Online) 17th March Morning Shift

If the fourth term in the expansion of $\left(x+x^{\log_2 x}\right)^7$ is 4480, then the value of x where x∈N is equal to :

(A) 3 (B) 1 (C) 4 (D) 2

Q76. JEE Main 2021 (Online) 16th March Morning Shift

If n is the number of irrational terms in the expansion of $\left(3^{1/4}+5^{1/8}\right)^{60}$, then (n − 1) is divisible by :

(A) 30 (B) 8 (C) 7 (D) 26

Q77. JEE Main 2021 (Online) 26th February Morning Shift

The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$ where x∈(0, 1) is :

(A) $\frac{10!}{\sqrt{3}(5!)^2}$ (B) $\frac{2.10!}{3\sqrt{3}(5!)^2}$

(C) $\frac{10!}{3(5!)^2}$ (D) $\frac{2.10!}{3(5!)^2}$

Q78. JEE Main 2020 (Online) 6th September Evening Slot

If the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{k}{x^2}\right)^{10}$ is 405, then |k| equals :

(A) 3 (B) 9 (C) 1 (D) 2

Q79. JEE Main 2020 (Online) 6th September Morning Slot

If {p} denotes the fractional part of the number p, then $\left\{\frac{3^{200}}{8}\right\}$, is equal to :

(A) $\frac{5}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) $\frac{3}{8}$

Q80. JEE Main 2020 (Online) 3rd September Evening Slot

If the term independent of x in the expansion of $\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is k, then 18 k is equal to :

(A) 5 (B) 9 (C) 7 (D) 11

Q81. JEE Main 2020 (Online) 3rd September Morning Slot

If the number of integral terms in the expansion of $(3^{1/2} + 5^{1/8})^n$ is exactly 33, then the least value of n is :

(A) 264 (B) 256 (C) 128 (D) 248

Q82. JEE Main 2020 (Online) 2nd September Morning Slot

Let $\alpha > 0$, $\beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}} + \beta x^{-\frac{1}{6}}\right)^{10}$ is 10k, then k is equal to :

(A) 176 (B) 336 (C) 352 (D) 84

Q83. JEE Main 2020 (Online) 9th January Evening Slot

In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$, if ℓ_1 is the least value of the term independent of x when $\frac{\pi}{8} \le \theta \le \frac{\pi}{4}$ and ℓ_2 is the least value of the term independent of x when $\frac{\pi}{16} \le \theta \le \frac{\pi}{8}$, then the ratio $\ell_2 : \ell_1$ is equal to :

(A) 8 : 1 (B) 16 : 1 (C) 1 : 8 (D) 1 : 16

Q84. JEE Main 2020 (Online) 8th January Evening Slot

If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x + \sqrt{x^2-1}\right)^6 + \left(x - \sqrt{x^2-1}\right)^6$, then

(A) $\alpha + \beta = 60$ (B) $\alpha - \beta = 60$ (C) $\alpha + \beta = -30$ (D) $\alpha - \beta = -132$

Q85. JEE Main 2020 (Online) 7th January Evening Slot

The coefficient of x^7 in the expression $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + ...+ x^{10}$ is:

(A) 120 (B) 330 (C) 420 (D) 210

Q86. JEE Main 2020 (Online) 7th January Morning Slot

The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + ... + 49^2 + 49 + 1$, is:

(A) 32 (B) 60 (C) 63 (D) 65

Q87. JEE Main 2019 (Online) 12th April Evening Slot

The term independent of x in the expansion of $\left(\frac{1}{60}-\frac{x^8}{81}\right).\left(2x^2-\frac{3}{x^2}\right)^6$ is equal to :

(A) 36 (B) - 108 (C) - 36 (D) - 72

Q88. JEE Main 2019 (Online) 12th April Morning Slot

The coefficient of x^{18} in the product $(1 + x)(1 - x)^{10}(1 + x + x^2)^9$ is :

(A) 126 (B) - 84 (C) - 126 (D) 84

Q89. JEE Main 2019 (Online) 10th April Evening Slot

The smallest natural number n, such that the coefficient of x in the expansion of $\left(x^2+\frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

(A) 23 (B) 58 (C) 38 (D) 35

Q90. JEE Main 2019 (Online) 9th April Morning Slot

If the fourth term in the binomial expansion of $\left(\frac{2}{x}+x^{\log_8 x}\right)^6$ (x > 0) is 20×8^7, then a value of x is :

(A) 8^{-2} (B) 8^2 (C) 8^3 (D) 8

Q91. JEE Main 2019 (Online) 8th April Evening Slot

If the fourth term in the binomial expansion of $\left(\sqrt{x^{\left(\frac{1}{1+\log_{10}x}\right)}}+x^{\frac{1}{12}}\right)^6$ is equal to 200, and x > 1, then the value of x is :

(A) 100 (B) 10^3 (C) 10 (D) 10^4

Q92. JEE Main 2019 (Online) 12th January Evening Slot

The total number of irrational terms in the binomial expansion of $(7^{1/5} - 3^{1/10})^{60}$ is :

(A) 54 (B) 55 (C) 49 (D) 48

Q93. JEE Main 2019 (Online) 12th January Morning Slot

A ratio of the 5th term from the beginning to the 5th term from the end in the binomial expansion of $\left(2^{1/3}+\frac{1}{2(3)^{1/3}}\right)^{10}$ is :

(A) $1 : 2(6)^{1/3}$ (B) $1 : 4(6)^{1/3}$ (C) $2(36)^{1/3} : 1$ (D) $4(36)^{1/3} : 1$

Q94. JEE Main 2019 (Online) 11th January Morning Slot

The sum of the real values of x for which the middle term in the binomial expansion of $\left(\frac{x^3}{3}+\frac{3}{x}\right)^8$ equals 5670 is :

(A) 0 (B) 8 (C) 6 (D) 4

Q95. JEE Main 2019 (Online) 10th January Evening Slot

If $\sum_{r=0}^{25}\left\{{}^{50}C_r.\ {}^{50-r}C_{25-r}\right\}=K\left({}^{50}C_{25}\right)$, then K is equal to :

(A) 2^{24} (B) $2^{25}-1$ (C) 2^{25} (D) $(25)^2$

Q96. JEE Main 2019 (Online) 10th January Evening Slot

The positive value of λ for which the co-efficient of x^2 in the expression $x^2\left(\sqrt{x}+\frac{\lambda}{x^2}\right)^{10}$ is 720, is -

(A) 4 (B) $2\sqrt{2}$ (C) 3 (D) $\sqrt{5}$

Q97. JEE Main 2019 (Online) 10th January Morning Slot

If the third term in the binomial expansion of $\left(1+x^{\log_2 x}\right)^5$ equals 2560, then a possible value of x is -

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

Q98. JEE Main 2019 (Online) 9th January Morning Slot

If the fractional part of the number $\left\{\frac{2^{403}}{15}\right\}$ is $\frac{k}{15}$, then k is equal to :

(A) 8 (B) 14 (C) 6 (D) 1

Q99. JEE Main 2017 (Online) 9th April Morning Slot

The coefficient of x^{-5} in the binomial expansion of

$\left(\frac{x+1}{x^{\frac{2}{3}}-x^{\frac{1}{3}}+1}-\frac{x-1}{x-x^{\frac{1}{2}}}\right)^{10}$, where $x\neq 0, 1$, is :

(A) 1 (B) 4 (C) -4 (D) -1

Q100. JEE Main 2017 (Online) 8th April Morning Slot

If $(27)^{999}$ is divided by 7, then the remainder is :

(A) 1 (B) 2 (C) 3 (D) 6

Q101. JEE Main 2016 (Online) 10th April Morning Slot

If the coefficients of x^{-2} and x^{-4} in the expansion of $\left(x^{\frac{1}{3}}+\frac{1}{2x^{\frac{1}{3}}}\right)^{18}, (x>0)$, are m and n respectively, then $\frac{m}{n}$ is equal to :

(A) 182 (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) 27

Q102. JEE Main 2016 (Online) 9th April Morning Slot

For $x \in \mathbf{R}$, $x \neq -1$,

if $(1+x)^{2016} + x(1+x)^{2015} + x^2(1+x)^{2014} + \ldots + x^{2016} = \sum_{i=0}^{2016} a_i x^i$, then a_{17} is equal to :

(A) $\frac{2017!}{17!\ 2000!}$ (B) $\frac{2016!}{17!\ 1999!}$ (C) $\frac{2017!}{2000!}$ (D) $\frac{2016!}{16!}$

Q103. JEE Main 2016 (Online) 9th April Morning Slot

The value of $\sum_{r=1}^{15} r^2 \left(\frac{^{15}C_r}{^{15}C_{r-1}}\right)$ is equal to :

(A) 560 (B) 680 (C) 1240 (D) 1085

Q104. JEE Main 2013 (Offline)

The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is

(A) 4 (B) 120 (C) 210 (D) 310

Q105. AIEEE 2012

If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is :

(A) an irrational number

(B) an odd positive integer

(C) an even positive integer

(D) a rational number other than positive integers

Q106. AIEEE 2009

The remainder left out when $8^{2n}-(62)^{2n+1}$ is divided by 9 is :

(A) 2 (B) 7 (C) 8 (D) 0

Q107. AIEEE 2007

In the binomial expansion of $(a-b)^n,\ \ n \geq 5$, the sum of 5th and 6th terms is zero, then a/b equals

(A) $\frac{n-5}{6}$ (B) $\frac{n-4}{5}$ (C) $\frac{5}{n-4}$ (D) $\frac{6}{n-5}$

Q108. AIEEE 2006

If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0 + a_1x + a_2x^2 + a_3x^3 \ldots\ldots$ then a_n is

(A) $\frac{b^n-a^n}{b-a}$ (B) $\frac{a^n-b^n}{b-a}$

(C) $\frac{a^{n+1}-b^{n+1}}{b-a}$ (D) $\frac{b^{n+1}-a^{n+1}}{b-a}$

Q109. AIEEE 2005

If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation

(A) $a-b=1$ (B) $a+b=1$ (C) $\frac{a}{b}=1$ (D) $ab=1$

Q110. AIEEE 2005

If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{\frac{3}{2}}-\left(1+\frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{2}}}$ may be approximated as

(A) $1-\frac{3}{8}x^2$ (B) $3x+\frac{3}{8}x^2$ (C) $-\frac{3}{8}x^2$ (D) $\frac{x}{2}-\frac{3}{8}x^2$

Q111. AIEEE 2005

If the coefficients of rth, (r+1)th, and (r + 2)th terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation

(A) $m^2 - m(4r-1) + 4r^2 - 2 = 0$ (B) $m^2 - m(4r+1) + 4r^2 + 2 = 0$

(C) $m^2 - m(4r+1) + 4r^2 - 2 = 0$ (D) $m^2 - m(4r-1) + 4r^2 + 2 = 0$

Q112. AIEEE 2004

The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and $(1-\alpha x)^6$ is the same if α equals

(A) $\frac{3}{5}$ (B) $\frac{10}{3}$ (C) $\frac{-3}{10}$ (D) $\frac{-5}{3}$

Q113. AIEEE 2004

The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is

(A) $(-1)^{n-1}n$ (B) $(-1)^n(1-n)$

(C) $(-1)^{n-1}(n-1)^2$ (D) $(n-1)$

Q114. AIEEE 2003

If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is

(A) 6th term (B) 7th term (C) 5th term (D) 8th term.

Q115. AIEEE 2003

The number of integral terms in the expansion of $\left(\sqrt{3}+\sqrt[8]{5}\right)^{256}$ is

(A) 35 (B) 32 (C) 33 (D) 34

Q116. AIEEE 2002

The coefficients of x^p and x^q in the expansion of $(1+x)^{p+q}$ are

(A) equal (B) equal with opposite signs

(C) reciprocals of each other (D) none of these

Q117. AIEEE 2002

The positive integer just greater than $(1+0.0001)^{10000}$ is

(A) 4 (B) 5 (C) 2 (D) 3

Numerical 68 questions

Q118. JEE Main 2026 (Online) 6th April Evening Shift

If $\left(1-x^3\right)^{10}=\sum_{r=0}^{10} a_r x^r(1-x)^{30-2r}$, then $\frac{9a_9}{a_{10}}$ is equal to ____ .

Q119. JEE Main 2026 (Online) 5th April Morning Shift

If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^3}-x^4\right)^n, x \neq 0$, is zero, then the value of n is ____ .

Q120. JEE Main 2026 (Online) 21st January Evening Shift

If $\left(\frac{1}{{}^{15}C_0}+\frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1}+\frac{1}{{}^{15}C_2}\right)\ldots\left(\frac{1}{{}^{15}C_{12}}+\frac{1}{{}^{15}C_{13}}\right)=\frac{\alpha^{13}}{{}^{14}C_0\,{}^{14}C_1\ldots{}^{14}C_{12}}$, then 30α is equal to ____ .

Q121. JEE Main 2025 (Online) 8th April Evening Shift

The product of the last two digits of $(1919)^{1919}$ is

Q122. JEE Main 2025 (Online) 7th April Evening Shift

The sum of the series
$2 \times 1 \times {}^{20}C_4 - 3 \times 2 \times {}^{20}C_5 + 4 \times 3 \times {}^{20}C_6 - 5 \times 4 \times {}^{20}C_7 + \cdots\cdots + 18 \times 17 \times {}^{20}C_{20}$, is equal to _____.

Q123. JEE Main 2025 (Online) 3rd April Evening Shift

Let $\left(1+x+x^2\right)^{10} = a_0 + a_1x + a_2x^2 + \ldots + a_{20}x^{20}$. If $(a_1 + a_3 + a_5 + \ldots + a_{19}) - 11a_2 = 121k$, then k is equal to_____ .

Q124. JEE Main 2025 (Online) 28th January Morning Shift

If $\alpha = 1 + \sum_{r=1}^{6}(-3)^{r-1}{}^{12}C_{2r-1}$, then the distance of the point $(12, \sqrt{3})$ from the line $\alpha x - \sqrt{3}y + 1 = 0$ is _____.

Q125. JEE Main 2025 (Online) 23rd January Morning Shift

The sum of all rational terms in the expansion of $\left(1 + 2^{1/3} + 3^{1/2}\right)^6$ is equal to _____.

Q126. JEE Main 2025 (Online) 22nd January Evening Shift

If $\sum_{r=1}^{30} \frac{r^2\left({}^{30}C_r\right)^2}{{}^{30}C_{r-1}} = \alpha \times 2^{29}$, then α is equal to _____.

Q127. JEE Main 2025 (Online) 22nd January Morning Shift

If $\sum_{r=0}^{5} \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{m} - \mathrm{n}$ is equal to _____.

Q128. JEE Main 2024 (Online) 9th April Morning Shift

The remainder when 428^{2024} is divided by 21 is _____.

Q129. JEE Main 2024 (Online) 8th April Morning Shift

Let $\alpha=\sum_{r=0}^{n}\left(4r^2+2r+1\right){}^nC_r$ and $\beta=\left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right)+\frac{1}{n+1}$. If $140<\frac{2\alpha}{\beta}<281$, then the value of n is _____.

Q130. JEE Main 2024 (Online) 6th April Morning Shift

If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3+x^2+y\right)$ is equal to _____.

Q131. JEE Main 2024 (Online) 5th April Morning Shift

If the constant term in the expansion of $\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is p, then $108\mathrm{p}$ is equal to _____.

Q132. JEE Main 2024 (Online) 4th April Morning Shift

Let $a=1+\frac{{}^2\mathrm{C}_2}{3!}+\frac{{}^3\mathrm{C}_2}{4!}+\frac{{}^4\mathrm{C}_2}{5!}+\ldots,\mathrm{b}=1+\frac{{}^1\mathrm{C}_0+{}^1\mathrm{C}_1}{1!}+\frac{{}^2\mathrm{C}_0+{}^2\mathrm{C}_1+{}^2\mathrm{C}_2}{2!}+\frac{{}^3\mathrm{C}_0+{}^3\mathrm{C}_1+{}^3\mathrm{C}_2+{}^3\mathrm{C}_3}{3!}+\ldots.$ Then $\frac{2b}{a^2}$ is equal to _____.

Q133. JEE Main 2024 (Online) 1st February Morning Shift

If the coefficient of x^{30} in the expansion of $\left(1+\frac{1}{x}\right)^6\left(1+x^2\right)^7\left(1-x^3\right)^8;x\neq0$ is α, then $|\alpha|$ equals _____.

Q134. JEE Main 2024 (Online) 31st January Evening Shift

Let the coefficient of x^r in the expansion of $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+\ldots\ldots\ldots+(x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r=\beta^n-\gamma^n,\beta,\gamma\in\mathbb{N}$, then the value of $\beta^2+\gamma^2$ equals _____.

Q135. JEE Main 2024 (Online) 31st January Morning Shift

In the expansion of $(1+x)\left(1-x^2\right)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5,x\neq0$, the sum of the coefficients of x^3 and x^{-13} is equal to _____.

Q136. JEE Main 2024 (Online) 30th January Evening Shift

Let $\alpha=\sum_{k=0}^{n}\left(\frac{({}^nC_k)^2}{k+1}\right)$ and $\beta=\sum_{k=0}^{n-1}\left(\frac{{}^nC_k{}^nC_{k+1}}{k+2}\right)$ If $5\alpha=6\beta$, then n equals _____.

Q137. JEE Main 2024 (Online) 30th January Morning Shift

Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____.

Q138. JEE Main 2024 (Online) 29th January Evening Shift

Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

Q139. JEE Main 2024 (Online) 29th January Morning Shift

If $\frac{{}^{11}C_1}{2}+\frac{{}^{11}C_2}{3}+\ldots+\frac{{}^{11}C_9}{10}=\frac{n}{m}$ with $\gcd(n,m)=1$, then $n+m$ is equal to _____.

Q140. JEE Main 2024 (Online) 27th January Evening Shift

The coefficient of x^{2012} in the expansion of $(1-x)^{2008}\left(1+x+x^2\right)^{2007}$ is equal to _____.

Q141. JEE Main 2023 (Online) 13th April Evening Shift

The remainder, when 7^{103} is divided by 17, is _____

Q142. JEE Main 2023 (Online) 13th April Morning Shift

Let α be the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{6}{x^{\frac{3}{2}}}\right)^n, n \leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

Q143. JEE Main 2023 (Online) 11th April Morning Shift

The mean of the coefficients of $x, x^2, \ldots, x^7$ in the binomial expansion of $(2+x)^9$ is _____.

Q144. JEE Main 2023 (Online) 11th April Morning Shift

The number of integral terms in the expansion of $\left(3^{\frac{1}{2}}+5^{\frac{1}{4}}\right)^{680}$ is equal to _____.

Q145. JEE Main 2023 (Online) 10th April Morning Shift

The coefficient of x^7 in $(1-x+2x^3)^{10}$ is _____.

Q146. JEE Main 2023 (Online) 8th April Morning Shift

Let $[t]$ denote the greatest integer $\leq t$. If the constant term in the expansion of $\left(3x^2-\frac{1}{2x^5}\right)^7$ is α, then $[\alpha]$ is equal to _____.

Q147. JEE Main 2023 (Online) 8th April Morning Shift

The largest natural number n such that 3^n divides $66!$ is _____.

Q148. JEE Main 2023 (Online) 1st February Evening Shift

Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ in the increasing powers of $2^{(x-2)\log_2 3}$, be 21 . If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____.

Q149. JEE Main 2023 (Online) 1st February Evening Shift

If the term without x in the expansion of $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ is 7315 , then $|\alpha|$ is equal to _____.

Q150. JEE Main 2023 (Online) 1st February Morning Shift

The remainder, when $19^{200}+23^{200}$ is divided by 49 , is _____.

Q151. JEE Main 2023 (Online) 31st January Evening Shift

The coefficient of x^{-6}, in the

expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is

Q152. JEE Main 2023 (Online) 31st January Evening Shift

If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^l}\right)^9$ is -84 and the coefficient of x^{-3l} is $2^\alpha\beta$, where $\beta<0$ is an odd number, then $|\alpha l-\beta|$ is equal to _____.

Q153. JEE Main 2023 (Online) 31st January Morning Shift

The remainder on dividing 5^{99} by 11 is _____.

Q154. JEE Main 2023 (Online) 31st January Morning Shift

Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ has a term $\beta x^{-\alpha}, \beta \in \mathbb{N}$. Then α is equal to _____.

Q155. JEE Main 2023 (Online) 30th January Evening Shift

50th root of a number x is 12 and 50th root of another number y is 18 . Then the remainder obtained on dividing $(x + y)$ by 25 is _____.

Q156. JEE Main 2023 (Online) 29th January Morning Shift

Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of those three terms, is _____.

Q157. JEE Main 2023 (Online) 29th January Morning Shift

If the co-efficient of x^9 in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to _____.

Q158. JEE Main 2023 (Online) 25th January Evening Shift

The remainder when $(2023)^{2023}$ is divided by 35 is _____.

Q159. JEE Main 2023 (Online) 25th January Morning Shift

The constant term in the expansion of $\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ is _____.

Q160. JEE Main 2023 (Online) 24th January Evening Shift

Let the sum of the coefficients of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n, x \neq 0. n \in \mathbb{N}$, be 376. Then the coefficient of x^4 is _____.

Q161. JEE Main 2022 (Online) 29th July Morning Shift

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^{n}$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}: 1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _____.

Q162. JEE Main 2022 (Online) 28th July Evening Shift

Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}}+\beta x\right)^{4}$, $(1-3\beta x)^{2}$ and $\left(1-\frac{\beta}{2}x\right)^{6}, \beta>0$, respectively form the first three terms of an A.P. If d is the common difference of this A.P. , then $50-\frac{2d}{\beta^{2}}$ is equal to _____.

Q163. JEE Main 2022 (Online) 27th July Evening Shift

Let for the 9th term in the binomial expansion of $(3+6x)^{n}$, in the increasing powers of $6x$, to be the greatest for $x=\frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k+n_0$ is equal to :

Q164. JEE Main 2022 (Online) 25th July Morning Shift

If the maximum value of the term independent of t in the expansion of $\left(t^{2}x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}, x \geqslant 0$, is K, then $8K$ is equal to _____.

Q165. JEE Main 2022 (Online) 29th June Evening Shift

Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}}-\frac{1}{x^{\frac{1}{5}}}\right)^{15}, x>0$, be m and n respectively. If r is a positive integer such that $mn^{2}={}^{15}C_{r}\cdot 2^{r}$, then the value of r is equal to _____.

Q166. JEE Main 2022 (Online) 28th June Morning Shift

The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^{3}+\frac{3}{x^{k}}\right)^{12}$, $x \neq 0$ is $2^{8}\cdot l$, where l is an odd integer, is _____.

Q167. JEE Main 2022 (Online) 27th June Morning Shift

If the coefficient of x^{10} in the binomial expansion of $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}}+\frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ is $5^{k}\cdot l$, where $l, k \in N$ and l is co-prime to 5, then k is equal to _____.

Q168. JEE Main 2022 (Online) 25th June Evening Shift

If the sum of the co-efficient of all the positive even powers of x in the binomial expansion of $\left(2x^3+\frac{3}{x}\right)^{10}$ is $5^{10}-\beta\,.\,3^9$, then β is equal to _____.

Q169. JEE Main 2022 (Online) 24th June Evening Shift

The remainder on dividing $1 + 3 + 3^2 + 3^3 + ... + 3^{2021}$ by 50 is _____.

Q170. JEE Main 2021 (Online) 31st August Evening Shift

If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to _____.

Q171. JEE Main 2021 (Online) 31st August Morning Shift

If $\left(\frac{3^6}{4^4}\right)k$ is the term, independent of x, in the binomial expansion of $\left(\frac{x}{4}-\frac{12}{x^2}\right)^{12}$, then k is equal to _____.

Q172. JEE Main 2021 (Online) 27th August Evening Shift

$3\times 7^{22} + 2\times 10^{22} - 44$ when divided by 18 leaves the remainder _____.

Q173. JEE Main 2021 (Online) 25th July Evening Shift

If the co-efficient of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n$ are equal, then the value of n is equal to _____.

Q174. JEE Main 2021 (Online) 25th July Morning Shift

The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1 + x)^{19}$ is _____.

Q175. JEE Main 2021 (Online) 25th July Morning Shift

The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to _____.

Q176. JEE Main 2021 (Online) 22th July Evening Shift

If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to _____.

Q177. JEE Main 2021 (Online) 20th July Morning Shift

The number of rational terms in the binomial expansion of $\left(4^{\frac{1}{4}} + 5^{\frac{1}{6}}\right)^{120}$ is _____.

Q178. JEE Main 2021 (Online) 18th March Evening Shift

The term independent of x in the expansion of

$\left[\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right]^{10}$, $x \neq 1$, is equal to _____.

Q179. JEE Main 2021 (Online) 17th March Evening Shift

Let the coefficients of third, fourth and fifth terms in the expansion of $\left(x + \frac{a}{x^2}\right)^n, x \neq 0$, be in the ratio 12 : 8 : 3. Then the term independent of x in the expansion, is equal to _____.

Q180. JEE Main 2021 (Online) 17th March Morning Shift

If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

Q181. JEE Main 2021 (Online) 25th February Evening Shift

If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is _____.

Q182. JEE Main 2021 (Online) 25th February Evening Shift

The total number of two digit numbers 'n', such that $3^n + 7^n$ is a multiple of 10, is _____.

Q183. JEE Main 2020 (Online) 5th September Morning Slot

The natural number m, for which the coefficient of x in the binomial expansion of

$\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is .....

Q184. JEE Main 2020 (Online) 4th September Morning Slot

Let $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20} a_r x^r$

Then $\frac{a_7}{a_{13}}$ is equal to _____.

Q185. JEE Main 2020 (Online) 9th January Morning Slot

The coefficient of x^4 is the expansion of $(1 + x + x^2)^{10}$ is _____.

Out of syllabus 70 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q186. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

If $\frac{1}{n+1}{}^n\mathrm{C}_n+\frac{1}{n}{}^n\mathrm{C}_{n-1}+\ldots+\frac{1}{2}{}^n\mathrm{C}_1+{}^n\mathrm{C}_0=\frac{1023}{10}$ then n is equal to :

(A) 9 (B) 6 (C) 7 (D) 8

Q187. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to

(A) ${}^{99}\mathrm{C}_{49}$ (B) ${}^{101}\mathrm{C}_{50}$ (C) $-{}^{99}\mathrm{C}_{49}$ (D) $-{}^{101}\mathrm{C}_{50}$

Q188. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

If the coefficients of three consecutive terms in the expansion of $(1+x)^n$ are in the ratio $1:5:20$, then the coefficient of the fourth term is

(A) 3654 (B) 1827 (C) 5481 (D) 2436

Q189. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

If ${}^{2n}C_3 : {}^nC_3 = 10:1$, then the ratio $\left(n^2+3n\right):\left(n^2-3n+4\right)$ is :

(A) $27:11$ (B) $2:1$ (C) $35:16$ (D) $65:37$

Q190. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

The coefficient of x^{301} in $(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+\ldots+x^{500}$ is :

(A) ${}^{500}C_{300}$ (B) ${}^{501}C_{200}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{302}$

Q191. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to

(A) 3025 (B) 4895 (C) 5445 (D) 1210

Q192. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ then α is equal to :

(A) 30 (B) 10 (C) 15 (D) 60

Q193. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{22} {}^{22}C_r\,{}^{23}C_r$ is

(A) ${}^{44}C_{23}$ (B) ${}^{45}C_{23}$ (C) ${}^{44}C_{22}$ (D) ${}^{45}C_{24}$

Q194. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

$\sum_{r=1}^{20} (r^2+1)(r!)$ is equal to

(A) $22! - 21!$ (B) $22! - 2(21!)$ (C) $21! - 2(20!)$ (D) $21! - 20!$

Q195. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i\,{}^nC_j$ is equal to

(A) $2^{2n} - {}^{2n}C_n$ (B) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(C) $2^{2n} - \frac{1}{2}{}^{2n}C_n$ (D) $2^{2n-1} + {}^{2n-1}C_n$

Q196. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let n $\geq$ 5 be an integer. If $9^n - 8n - 1 = 64\alpha$ and $6^n - 5n - 1 = 25\beta$, then $\alpha - \beta$ is equal to

(A) $1 + {}^nC_2(8-5) + {}^nC_3(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-1}-5^{n-1})$

(B) $1 + {}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-2}-5^{n-2})$

(C) ${}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-2}-5^{n-2})$

(D) ${}^nC_4(8-5) + {}^nC_5(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-3}-5^{n-3})$

Q197. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

If

$$\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right)-\sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right)=\frac{\alpha(60!)}{(30!)(31!)},$$

where $\alpha \in$ R, then the value of 16α is equal to

(A) 1411 (B) 1320 (C) 1615 (D) 1855

Q198. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

$\sum_{k=0}^{20}\left({}^{20}C_k\right)^2$ is equal to :

(A) ${}^{40}C_{21}$ (B) ${}^{40}C_{19}$ (C) ${}^{40}C_{20}$ (D) ${}^{41}C_{20}$

Q199. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1 + x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 . {}^{20}C_r$ is equal to :

(A) 420×2^{19} (B) 380×2^{19} (C) 380×2^{18} (D) 420×2^{18}

Q200. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

For the natural numbers m, n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \ldots . + a_{m+n}y^{m+n}$ and $a_1 = a_2 = 10$, then the value of (m + n) is equal to :

(A) 88 (B) 64 (C) 100 (D) 80

Q201. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

The coefficient of x^{256} in the expansion of

$(1 - x)^{101} (x^2 + x + 1)^{100}$ is :

(A) ${}^{100}C_{16}$ (B) ${}^{100}C_{15}$ (C) $-{}^{100}C_{16}$ (D) $-{}^{100}C_{15}$

Q202. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

Let $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + ... + a_{40}x^{40}$. Then $a_1 + a_3 + a_5 + ... + a_{37}$ is equal to

(A) $2^{20}(2^{20} - 21)$ (B) $2^{19}(2^{20} - 21)$ (C) $2^{19}(2^{20} + 21)$ (D) $2^{20}(2^{20} + 21)$

Q203. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{6}\left({}^6C_r . {}^6C_{6-r}\right)$ is equal to :

(A) 924 (B) 1024 (C) 1124 (D) 1324

Q204. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Let [x] denote greatest integer less than or equal to x. If for n∈N,

$(1-x+x^3)^n=\sum_{j=0}^{3n} a_j x^j$,

then $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j}+4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1}$ is equal to :

(A) 2^n-1 (B) n (C) 2 (D) 1

Q205. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

If $n \geq 2$ is a positive integer, then the sum of the series $^{n+1}C_2+2\left(^2C_2+^3C_2+^4C_2+\ldots+^nC_2\right)$ is :

(A) $\frac{n(2n+1)(3n+1)}{6}$ (B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(n+1)^2(n+2)}{12}$ (D) $\frac{n(n-1)(2n+1)}{6}$

Q206. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The value of

$-{}^{15}C_1+2.{}^{15}C_2-3.{}^{15}C_3+\ldots-15.{}^{15}C_{15}+{}^{14}C_1+{}^{14}C_3+{}^{14}C_5+\ldots+{}^{14}C_{11}$ is :

(A) $2^{13}-13$ (B) $2^{16}-1$ (C) 2^{14} (D) $2^{13}-14$

Q207. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

If for some positive integer n, the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14, then the largest coefficient in this expansion is :

(A) 330 (B) 792 (C) 252 (D) 462

Q208. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to:

(A) $^{50}C_6-{}^{30}C_6$ (B) $^{51}C_7-{}^{30}C_7$ (C) $^{50}C_7-{}^{30}C_7$ (D) $^{51}C_7+{}^{30}C_7$

Q209. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

If $^{20}C_1+(2^2)\,{}^{20}C_2+(3^2)\,{}^{20}C_3+\ldots..+(20^2)\,{}^{20}C_{20}=A(2^{\beta})$, then the ordered pair (A, β) is equal to :

(A) (420, 19) (B) (420, 18) (C) (380, 18) (D) (380, 19)

Q210. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x, then the ordered pair (a,b) is equal to :

(A) (28, 861) (B) (28, 315) (C) (−21, 714) (D) (−54, 315)

Q211. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

If some three consecutive coefficients in the binomial expansion of $(x + 1)^n$ in powers of x are in the ratio 2 : 15 : 70, then the average of these three coefficients is :-

(A) 625 (B) 227 (C) 964 (D) 232

Q212. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The sum of the series

$2.{}^{20}C_0 + 5.{}^{20}C_1 + 8.{}^{20}C_2 + 11.{}^{20}C_3 + ... + 62.{}^{20}C_{20}$ is equal to :

(A) 2^{25} (B) 2^{24} (C) 2^{26} (D) 2^{23}

Q213. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The sum of the co-efficients of all even degree terms in x in the expansion of $\left(x+\sqrt{x^3-1}\right)^6 + \left(x-\sqrt{x^3-1}\right)^6$, (x > 1) is equal to:

(A) 32 (B) 26 (C) 29 (D) 24

Q214. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Let $(x + 10)^{50} + (x - 10)^{50} = a_0 + a_1x + a_2x^2 + ... + a_{50}x^{50}$, for all $x \in R$; then $\frac{a_2}{a_0}$ is equal to

(A) 12.25 (B) 12.75 (C) 12.00 (D) 12.50

Q215. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Let $S_n = 1 + q + q^2 + ... + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + ... + \left(\frac{q+1}{2}\right)^n$ where q is a real number and $q \neq 1$. If ${}^{101}C_1 + {}^{101}C_2 . S_1 + ... + {}^{101}C_{101} . S_{100} = \alpha T_{100}$ then α is equal to

(A) 202 (B) 200 (C) 2^{100} (D) 2^{99}

Q216. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The value of r for which ${}^{20}C_r\ {}^{20}C_0 + {}^{20}C_{r-1}\ {}^{20}C_1 + {}^{20}C_{r-2}\ {}^{20}C_2 + ... + {}^{20}C_0\ {}^{20}C_r$ is maximum, is

(A) 20 (B) 15 (C) 10 (D) 11

Q217. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

If $\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i+{}^{20}C_{i-1}}\right)^3=\frac{k}{21}$ then k is equal to

(A) 100 (B) 200 (C) 50 (D) 400

Q218. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is :

(A) 14 (B) 15 (C) 10 (D) 12

Q219. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The coefficient of x^2 in the expansion of the product
$(2-x^2).((1+2x+3x^2)^6+(1-4x^2)^6)$ is :

(A) 107 (B) 106 (C) 108 (D) 155

Q220. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

The sum of the co-efficients of all odd degree terms in the expansion of

$\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, (x>1)$ is

(A) 2 (B) -1 (C) 0 (D) 1

Q221. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

The coefficient of x^{10} in the expansion of $(1+x)^2(1+x^2)^3(1+x^3)^4$ is equal to :

(A) 52 (B) 56 (C) 50 (D) 44

Q222. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

If n is the degree of the polynomial,

$$\left[\frac{2}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\right]^8+\left[\frac{2}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\right]^8$$

and m is the coefficient of x^n in it, then the ordered pair (n, m) is equal to :

(A) $(24, (10)^8)$ (B) $(8, 5(10)^4)$ (C) $(12, (20)^4)$ (D) $(12, 8(10)^4)$

Q223. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The value of $\left({}^{21}C_1-{}^{10}C_1\right)+\left({}^{21}C_2-{}^{10}C_2\right)+\left({}^{21}C_3-{}^{10}C_3\right)$
$\left({}^{21}C_4-{}^{10}C_4\right)+\ldots.+\left({}^{21}C_{10}-{}^{10}C_{10}\right)$ is

(A) $2^{21}-2^{10}$ (B) $2^{20}-2^9$ (C) $2^{20}-2^{10}$ (D) $2^{21}-2^{11}$

Q224. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is :

(A) 243 (B) 729 (C) 64 (D) 2187

Q225. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The sum of coefficients of integral power of x in the binomial expansion $\left(1-2\sqrt{x}\right)^{50}$ is :

(A) $\frac{1}{2}\left(3^{50}-1\right)$ (B) $\frac{1}{2}\left(2^{50}+1\right)$

(C) $\frac{1}{2}\left(3^{50}+1\right)$ (D) $\frac{1}{2}\left(3^{50}\right)$

Q226. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If the coefficints of x^3 and x^4 in the expansion of $\left(1+ax+bx^2\right)(1-2x)^{18}$ in powers of x are both zero, then (a, b) is equal to:

(A) $\left(14, \frac{272}{3}\right)$ (B) $\left(16, \frac{272}{3}\right)$

(C) $\left(16, \frac{251}{3}\right)$ (D) $\left(14, \frac{251}{3}\right)$

Q227. AIEEE 2011 · MCQ (Single Correct Answer)

The coefficient of x^7 in the expansion of $\left(1-x-x^2+x^3\right)^6$ is

(A) -132 (B) -144 (C) 132 (D) 144

Q228. AIEEE 2010 · MCQ (Single Correct Answer)

Let $s_1=\sum_{j=1}^{10} j\,(j-1)^{10}C_j$,

$s_2=\sum_{j=1}^{10} j.\,^{10}C_j$ and

$s_3=\sum_{j=1}^{10} j^2.\,^{10}C_j$.

Statement-1 : $S_3=55\times 2^9$.

Statement-2 : $S_1=90\times 2^8$ and $S_2=10\times 2^8$.

(A) Statement - 1 is true, Statement- 2 is true; Statement - 2 is not a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement-2 is false.

(C) Statement - 1 is false, Statement-2 is true.

(D) Statement - 1 is true, Statement-2 is true; Statement - 2 is a correct explanation for Statement - 1.

Q229. AIEEE 2008 · MCQ (Single Correct Answer)

Statement - 1 : $\sum_{r=0}^{n}(r+1)\,{}^{n}C_{r}=(n+2)2^{n-1}$.

Statement - 2 : $\sum_{r=0}^{n}(r+1)\,{}^{n}C_{r}x^{r}=(1+x)^{n}+nx(1+x)^{n-1}$.

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Q230. AIEEE 2007 · MCQ (Single Correct Answer)

The sum of the series ${}^{20}C_{0}-{}^{20}C_{1}+{}^{20}C_{2}-{}^{20}C_{3}+\ldots\ldots-\ldots\ldots+{}^{20}C_{10}$ is

(A) 0 (B) ${}^{20}C_{10}$ (C) $-{}^{20}C_{10}$ (D) $\frac{1}{2}{}^{20}C_{10}$

Q231. AIEEE 2006 · MCQ (Single Correct Answer)

For natural numbers m , n, if $(1-y)^{m}(1+y)^{n}=1+a_{1}y+a_{2}y^{2}+\ldots\ldots\ldots$ and $a_{1}=a_{2}=10$, then $(m,\,n)$ is

(A) $(20,\,45)$ (B) $(35,\,20)$ (C) $(45,\,35)$ (D) $(35,\,45)$

Q232. AIEEE 2005 · MCQ (Single Correct Answer)

The value of ${}^{50}C_{4}+\sum_{r=1}^{6}{}^{56-r}C_{3}$ is

(A) ${}^{55}C_{4}$ (B) ${}^{55}C_{3}$ (C) ${}^{56}C_{3}$ (D) ${}^{56}C_{4}$

Q233. AIEEE 2004 · MCQ (Single Correct Answer)

If $S_{n}=\sum_{r=0}^{n}\frac{1}{{}^{n}C_{r}}\ and\ t_{n}=\sum_{r=0}^{n}\frac{r}{{}^{n}C_{r}}$, then $\frac{t_{n}}{S_{n}}$ is equal to

(A) $\frac{2n-1}{2}$ (B) $\frac{1}{2}n-1$ (C) n - 1 (D) $\frac{1}{2}n$

Q234. AIEEE 2002 · MCQ (Single Correct Answer)

If the sum of the coefficients in the expansion of $(a+b)^{n}$ is 4096, then the greatest coefficient in the expansion is

(A) 1594 (B) 792 (C) 924 (D) 2924

Q235. AIEEE 2002 · MCQ (Single Correct Answer)

r and n are positive integers $r > 1$, $n > 2$ and coefficient of $(r+2)^{th}$ term and $3r^{th}$ term in the expansion of $(1+x)^{2n}$ are equal, then n equals

(A) $3r$ (B) $3r+1$ (C) $2r$ (D) $2r+1$

Q236. JEE Main 2023 (Online) 6th April Morning Shift · Numerical

The coefficient of x^{18} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____.

Q237. JEE Main 2023 (Online) 24th January Morning Shift · Numerical

Suppose $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is _____

Q238. JEE Main 2022 (Online) 29th July Evening Shift · Numerical

If $\sum_{k=1}^{10} k^2 \left({}^{10}C_k\right)^2 = 22000L$, then L is equal to _____.

Q239. JEE Main 2022 (Online) 28th July Evening Shift · Numerical

If $1 + (2 + {}^{49}C_1 + {}^{49}C_2 + \ldots + {}^{49}C_{49})({}^{50}C_2 + {}^{50}C_4 + \ldots + {}^{50}C_{50})$ is equal to $2^{\mathrm{n}} \cdot \mathrm{m}$, where m is odd, then $\mathrm{n} + \mathrm{m}$ is equal to _____.

Q240. JEE Main 2022 (Online) 26th July Morning Shift · Numerical

If the coefficients of x and x^2 in the expansion of $(1+x)^{\mathrm{p}}(1-x)^{\mathrm{q}}, \mathrm{p}, \mathrm{q} \leq 15$, are -3 and -5 respectively, then the coefficient of x^3 is equal to _____.

Q241. JEE Main 2022 (Online) 27th June Evening Shift · Numerical

If the sum of the coefficients of all the positive powers of x, in the Binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral values of n is _____.

Q242. JEE Main 2022 (Online) 26th June Evening Shift · Numerical

If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots\ldots + \left({}^{60}C_{20}\right) = \frac{m}{n}\,{}^{60}C_{20}$ m and n are coprime, then m + n is equal to _____.

Q243. JEE Main 2022 (Online) 25th June Morning Shift · Numerical

Let C_r denote the binomial coefficient of x^r in the expansion of $(1+x)^{10}$. If for $\alpha, \beta \in R$, $C_1 + 3.2C_2 + 5.3C_3 + \ldots\ldots$ upto 10 terms $= \frac{\alpha \times 2^{11}}{2^{\beta}-1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots\ldots \; upto\, 10\, terms\right)$ then the value of $\alpha + \beta$ is equal to _____.

Q244. JEE Main 2021 (Online) 1st September Evening Shift · Numerical

If the sum of the coefficients in the expansion of $(x + y)^n$ is 4096, then the greatest coefficient in the expansion is _____.

Q245. JEE Main 2021 (Online) 26th August Evening Shift · Numerical

Let $\binom{n}{k}$ denotes ${}^nC_k$ and $\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \binom{n}{k}, & if\, 0 \le k \le n \\ 0, & otherwise \end{cases}$

If $A_k = \sum_{i=0}^{9} \binom{9}{i} \begin{bmatrix} 12 \\ 12-k+i \end{bmatrix} + \sum_{i=0}^{8} \binom{8}{i} \begin{bmatrix} 13 \\ 13-k+i \end{bmatrix}$ and $A_4 - A_3 = 190$ p, then p is equal to :

Q246. JEE Main 2021 (Online) 25th July Evening Shift · Numerical

Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of (n + 1) terms ${}^nC_0, 3.{}^nC_1, 5.{}^nC_2, 7.{}^nC_3, \ldots\ldots$ is equal to $2^{100} \cdot 101$, then $2\left[\frac{n-1}{2}\right]$ is equal to _____.

Q247. JEE Main 2021 (Online) 22th July Evening Shift · Numerical

The number of elements in the set $\{n \in \{1, 2, 3, \ldots\ldots, 100\} \mid (11)^n > (10)^n + (9)^n\}$ is _____.

Q248. JEE Main 2021 (Online) 18th March Evening Shift · Numerical

Let ${}^nC_r$ denote the binomial coefficient of x^r in the expansion of $(1 + x)^n$. If $\sum_{k=0}^{10} (2^2 + 3k)^{10} C_k = \alpha.3^{10} + \beta.2^{10}, \alpha, \beta \in R$, then $\alpha + \beta$ is equal to _____.

Q249. JEE Main 2021 (Online) 16th March Evening Shift · Numerical

Let n be a positive integer. Let

$A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$. If

$63A = 1 - \frac{1}{2^{30}}$, then n is equal to _____.

Q250. JEE Main 2021 (Online) 26th February Morning Shift · **Numerical**

Let m, n∈N and gcd (2, n) = 1. If $30\binom{30}{0}+29\binom{30}{1}+\ldots\ldots+2\binom{30}{28}+1\binom{30}{29}=n.2^m$, then n + m is equal to _____.

(Here $\binom{n}{k}={}^nC_k$)

Q251. JEE Main 2021 (Online) 24th February Evening Shift · **Numerical**

For integers n and r, let $\binom{n}{r}=\begin{cases}{}^nC_r, & if\ n\geq r\geq 0\\ 0, & otherwise\end{cases}$ The maximum value of k for which the sum $\sum_{i=0}^{k}\binom{10}{i}\binom{15}{k-i}+\sum_{i=0}^{k+1}\binom{12}{i}\binom{13}{k+1-i}$ exists, is equal to _____.

Q252. JEE Main 2020 (Online) 5th September Evening Slot · **Numerical**

The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^6$ in powers of x, is _____.

Q253. JEE Main 2020 (Online) 2nd September Evening Slot · **Numerical**

For a positive integer n,
$\left(1+\frac{1}{x}\right)^n$ is expanded
in increasing powers of x. If three consecutive coefficients in this expansion are in the ratio, 2 : 5 : 12, then n is equal to_____.

Q254. JEE Main 2020 (Online) 9th January Evening Slot · **Numerical**

If $C_r \equiv {}^{25}C_r$ and

$C_0 + 5.C_1 + 9.C_2 + ... + (101).C_{25} = 2^{25}.k$, then k is equal to _____.

Q255. JEE Main 2020 (Online) 7th January Morning Slot · **Numerical**

If the sum of the coefficients of all even powers of x in the product $(1 + x + x^2 + ...+ x^{2n})(1 - x + x^2 - x^3 + ... + x^{2n})$ is 61, then n is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **B**	4. **A**	5. **B**
6. **C**	7. **C**	8. **D**	9. **C**	10. **B**
11. **B**	12. **A**	13. **B**	14. **D**	15. **B**
16. **D**	17. **B**	18. **D**	19. **B**	20. **B**
21. **B**	22. **B**	23. **A**	24. **D**	25. **C**
26. **A**	27. **B**	28. **D**	29. **B**	30. **C**
31. **C**	32. **D**	33. **B**	34. **B**	35. **C**
36. **D**	37. **C**	38. **D**	39. **B**	40. *Scan QR*
41. **B**	42. **D**	43. **D**	44. **B**	45. **C**
46. **A**	47. **A**	48. **D**	49. **D**	50. **A**
51. **D**	52. **C**	53. **D**	54. **D**	55. **B**
56. **C**	57. **B**	58. **A**	59. **D**	60. **C**
61. **A**	62. **D**	63. **C**	64. **D**	65. **B**
66. **C**	67. **A**	68. **D**	69. **D**	70. **C**
71. **D**	72. **D**	73. **A**	74. **C**	75. **D**
76. **D**	77. **B**	78. **A**	79. **C**	80. **C**
81. **B**	82. **B**	83. **B**	84. **D**	85. **B**
86. **C**	87. **C**	88. **D**	89. **C**	90. **B**
91. **C**	92. **A**	93. **D**	94. **A**	95. **C**
96. **A**	97. **D**	98. **A**	99. **A**	100. **D**
101. **A**	102. **A**	103. **B**	104. **C**	105. **A**
106. **A**	107. **B**	108. **D**	109. **D**	110. **C**
111. **C**	112. **C**	113. **B**	114. **D**	115. **C**
116. **A**	117. **D**			

Numerical

118. **30**	119. **21**	120. **32**	121. **63**	122. **34**
123. **239**	124. **5**	125. **612**	126. **465**	127. **2035**
128. **1**	129. **5**	130. **806**	131. **54**	132. **8**
133. **678**	134. **25**	135. **118**	136. **10**	137. **138**
138. **1**	139. **2041**	140. **0**	141. **12**	142. **36**

143. **2736**	144. **171**	145. **960**	146. **1275**	147. **31**
148. **4**	149. **1**	150. **29**	151. **5040**	152. **98**
153. **9**	154. **2**	155. **23**	156. **1120**	157. **1**
158. **7**	159. **1080**	160. **405**	161. **84**	162. **57**
163. **24**	164. **6006**	165. **5**	166. **2**	167. **5**
168. **83**	169. **4**	170. **315**	171. **55**	172. **15**
173. **55**	174. **1**	175. **210**	176. **8**	177. **21**
178. **210**	179. **4**	180. **4**	181. **1**	182. **45**
183. **13**	184. **8**	185. **615**		

Out of syllabus

186. **A**	187. **C**	188. **A**	189. **B**	190. **B**
191. **D**	192. **C**	193. **B**	194. **B**	195. **A**
196. **C**	197. **A**	198. **C**	199. **D**	200. **D**
201. **B**	202. **B**	203. **A**	204. **D**	205. **B**
206. **D**	207. **D**	208. **B**	209. **B**	210. **B**
211. **D**	212. **A**	213. **D**	214. **A**	215. **C**
216. **A**	217. **A**	218. **B**	219. **B**	220. **A**
221. **A**	222. **C**	223. **C**	224. **B**	225. **C**
226. **B**	227. **B**	228. **B**	229. **B**	230. **D**
231. **D**	232. **D**	233. **D**	234. **C**	235. **C**
236. **5005**	237. **1012**	238. **221**	239. **99**	240. **23**
241. **57**	242. **102**	243. **286**	244. **924**	245. **49**
246. **98**	247. **96**	248. **19**	249. **6**	250. **45**
251. **12**	252. **120**	253. **118**	254. **51**	255. **30**